# Types of Environment

1. Fully observable / Partially Observable
2. Deterministic / Stochastic
3. Episodic / Sequential
4. Single-agent / Multi-agent
5. Static / Dynamic
6. Discrete / Continuous

# Fully observable / Partially Observable

## Fully observable

- An agent can always see the entire state of an environment.
- **Example**: Chess

## Partially Observable

- An agent can never see the entire state of an environment.
- **Example**: Card game

# Deterministic / Stochastic

## Deterministic

- An agent's current state and selected action can completely determine the next state of the environment.
- Example: Tic tac toe

## Stochastic

- A stochastic environment is random in nature and cannot be determined completely by an agent.
- Example: Ludo ( Any games that involve dice )

# Episodic / Sequential

## Episodic

- Only the current percept is required for the action
- Every episode is independent of each other
- **Example**: part picking robot

## Sequential

- An agent requires memory of past actions to determine the next best actions.
- The current decision could affect all future decisions.
- Example: Chess

# Single-agent / Multi-agent

## Single-agent

- If only one agent is involved in an environment, and operating by itself then such an environment is called single agent environment.
- **Example**: maze


## Multi-agent

- if multiple agents are operating in an environment, then such an environment is called a multi-agent environment.
- **Example**: football

# Static / Dynamic

## Static

- The environment does not change while an agent is acting
- **Example**: Crossword puzzles

## Dynamic

- The environment may change over time
- Example: roller coaster ride

  Taxi driving

# Discrete / Continuous

**Discrete**

- The environment consists of a finite number of actions that can be deliberated in the environment to obtain the output.
- **Example**: chess

**Continuous**

- The environment in which the actions performed cannot be numbered ie.,is not discrete, is said to be continuous.
- **Example**: Self-driving cars

# Known / Unknown

**Known**

- In a known environment, the results for all actions are known to the agent.
- Example: Card game

**Unknown**

- In unknown environment, agent needs to learn how it works in order to perform an action.
- Example: A new video game